Chapter 3 Intro to Networking Exam

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Score: 99% Passing Score: 80%

Question 1: ✔ Correct

Which of the following hardware devices links multiple networks and directs traffic between networks?

- Bridge
- Repeater
- Hub
- ➡ Router

EXPLANATION

A router is a device that links multiple networks and directs traffic between networks. Each network linked by routers has its own unique identifier called the network number or network address.

REFERENCES

LabSim for Network Pro, Section 3.3.

Question 2: ✔ Correct

An eight-port switch receives a frame on port number 1. The frame is addressed to an unknown device. What will the switch do?

- Send the frame out all eight ports.
- Send the frame out the destination port.
- ➡ Send the frame out ports two through eight.
- Drop the frame.

EXPLANATION

Because the switch does not know the port that is used to reach the destination device, it will send the frame out all ports except for the port on which the frame was received. After the switch learns the port that is used to reach the destination device, it will send the frame out only that port.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 3: ✔ Correct

At which OSI layer does a router operate to forward network messages?

- Data Link
- Session
- ➡ Network
- Transport
- Physical

EXPLANATION

A router uses the logical network address specified at the Network layer to forward messages to the appropriate LAN segment. A bridge, on the other hand, uses the MAC address and works at the Data Link layer.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 4: ✔ Correct

How do switches and bridges learn where devices are located on a network?

- When a frame enters a port, the source IP address is copied from the frame header.
- When a frame enters a port, the destination MAC address is copied from frame header.
- ➡ When a frame enters a port, the source MAC address is copied from the frame header.
- When a frame enters a port, the destination IP address is copied from the frame header.

EXPLANATION

Bridges and switches learn addresses by copying the MAC address of the source device and placing it into the MAC address table. The port number that the frame entered is also recorded in the table and associated with the source MAC address.

The switch or the bridge cannot record the destination MAC address because it does not know the port that is used to reach the destination device. Bridges and switches operate at Layer 2 and do not use IP addresses (which exist at Layer 3).

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 5: ✔ Correct

Which of the following hardware devices regenerates a signal out all connected ports without examining the frame or packet contents? (Select two.)

- [ ] Router
- [ ] Switch
- [x] Repeater
- [ ] Bridge
- [x] Hub
- [ ] Gateway

EXPLANATION

A hub and a repeater send received signals out all other ports. These devices do not examine the frame or the packet contents.

Switches and bridges use the MAC address in a frame for forwarding decisions. A router uses the IP address in a packet for forwarding decisions.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 6: ✔ Correct

Which of the following devices operate at the Data Link layer of the OSI model? (Select three.)

- [x] Bridges
- [ ] Routers
- [ ] Repeaters
- [x] Network interface cards (NICs)
- [ ] Hubs
- [x] Switches

EXPLANATION

Network interface cards (NICs), bridges, and switches all operate at the OSI Data Link layer. They use the physical device address (MAC address) to identify packets. Hubs and repeaters operate at the Physical layer--they simply repeat packets without regard to addresses. Routers function at the Network layer--they examine the logical device and network address to perform routing tasks.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 7: ✔ Correct

Which of the following devices operate at OSI model Layer 2? (Select two.)

- [ ] Router
- [x] Switch
- [ ] Repeater
- [ ] Hub
- [x] Network interface card
- [ ] Firewall

EXPLANATION

A network interface card and a switch operate at Layer 2 (Data Link) of the OSI model. Layer 2 includes protocols that define the MAC address. The MAC address is burned into the network interface card, and a switch uses the MAC address to make forwarding decisions.

A hub or a repeater operate at Layer 1; they regenerate a signal without looking at Layer 2 or Layer 3 information. A router operates at Layer 3, using the IP address to make forwarding decisions. A firewall operates at Layer 3 or higher, using packet or data contents to make filtering decisions.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 8: ✔ Correct

Which of the following is a valid MAC address?

- 73-99-12-61-15
- 255.255.255.0
- 192.168.12.15
- 34-9A-86-1G-B3-24
- 83-5A-5B-0B-31-55-F1
- ➡ C0-34-FF-15-01-8E

EXPLANATION

A MAC or hardware address is a unique identifier hard coded on every network adapter card. A valid MAC address has a total of 12 hexadecimal numbers. Hexadecimal numbers contain the numbers 0 to 9 and the letters A to F. Valid values for each octet in a MAC address range anywhere from 00 to FF. Note that one of the answers would be a valid MAC address except it uses a G value, which is beyond the hexadecimal range.

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 9: ✔ Correct

A switch is associated with which OSI model layer?

- Network
- ➡ Data Link
- Transport
- Physical

EXPLANATION

Switches are associated with the Data Link layer of the OSI model. Switches examine the device address in the packet and forward messages directly to that device.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 10: ✔ Correct

Which of the following statements accurately describes how a modem works? (Select two.)

- [ ] It demodulates analog PC data into digital data that can be transmitted through a telephone network.
- [ ] It modulates digital data from a telephone network into analog data that a PC can use.
- [x] It modulates digital data from the PC into analog data and transmits it on a telephone network.
- [x] It demodulates analog data from a telephone network into digital PC data.
- [ ] It transmits digital signals over ordinary telephone copper wiring at a rate up to 128 Kbps.
- [ ] It communicates over a telephone network using digital signals.

EXPLANATION

Modem is shorthand for modulator/demodulator. A modem's job is to convert (or modulate) digital data from a PC into analog telephone signals and transmit them through a telephone network.

Modems also receive analog data from the telephone network and convert (or demodulate) it into digital PC data.

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 11: — Partially Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You did not complete the lab correctly.

View Lab Report

Question 12: ✔ Correct

You want a switch to have the ability to modify the media type the switch port supports. Which type of module might you use to make this possible?

- ➡ ☑ GBIC
- ☐ OC-x
- ☐ MPLS
- ➡ ☑ SFP

EXPLANATION

Older network adapters used an external transceiver that matched the media type. While nearly all current network adapters come with a built-in transceiver type, new devices, such as switches and routers, use transceiver modules that allow you to modify the media type of a port by changing the transceiver. There are several types of transceiver modules.

- A GBIC (gigabit interface converter) is a large transceiver that fits in a port slot. GBICs are used for Gigabit media, including copper and fiber optic.
- An SFP (small form-factor pluggable) is similar to a GBIC, but is smaller in size. An SFP is sometimes called a mini-GBIC.
- An XFP transceiver is similar in size to an SFP, but is used for 10-Gigabit networking.

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 13: ✔ Correct

You have a server that has a 100BaseFX network interface card that you need to connect to a switch. The switch only has 100BaseTX switch ports.

Which device should you use?

- ➡ Media converter
- Hub
- Bridge
- Repeater
- Gateway

EXPLANATION

Use a media converter to convert from one media type to another media type within the same architecture.

Use a bridge to connect two devices that use different network architectures. For example, you can use a bridge to connect a wired network to wireless clients. A hub or a repeater connects devices using the same media type.

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 14: ✔ Correct

Which of the following is a valid MAC address?

- ➡ AB.07.CF.62.16.BD
- 95ABC2F4.ABC5.569D.43BF
- 145.65.254.10
- FABC.875E.9BG6

EXPLANATION

MAC addresses are comprised of 12 hexadecimal digits (ranging from 0-9 and A-F). They are typically represented as a three sets of four hexadecimal digits or six sets of two hexadecimal digits separated with periods. Regardless of the grouping and separator values, look for 12 hex digits to determine whether a MAC address is valid.

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 15: ✔ Correct

Which of the following devices operates at the OSI model Layer 1?

- ➡ Hub
- Switch
- Firewall
- Router

EXPLANATION

A hub operates at OSI model Layer 1 (Physical layer). It regenerates electrical signals and sends those signals out all hub ports without regard to the upper-layer data.

A switch operates at the OSI model Layer 2 (Data Link layer). It reads the MAC address to make frame forwarding decisions. A router and a firewall operate at OSI model Layer 3 (Network layer).

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 16: ✔ Correct

Which of the following is true about the MAC address? (Select two.)

- [x] It is typically represented by hexadecimal numbers.
- [x] It is a 48-bit address.
- [ ] It is a 64-bit address.
- [ ] It is typically represented by octets of decimal numbers between 0-255.
- [ ] It is a 32-bit address.

EXPLANATION

The MAC address identifies the physical address of the network adapter. The MAC address is a 12-digit (48-bit) hexadecimal number (each number ranges from 0-9 or A-F). The address is often written as 00-B0-D0-06-BC-AC or 00B0.D006.BCAC, although dashes, periods, and colons can be used to divide the MAC address parts.

An IPv4 address is 32-bits and uses octets of decimal numbers between 0-255. An IPv6 address is a 64-bit address that uses 32 hexadecimal numbers.

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 17: ✔ Correct

Which of the following devices operates at the OSI model Layer 2?

- ➡ Switch
- Firewall
- Router
- Hub

EXPLANATION

A switch operates at the OSI model Layer 2 (Data Link layer). It reads the MAC address to make frame forwarding decisions.

A hub operates at OSI model Layer 1 (Physical layer), while a router and a firewall operate at OSI model Layer 3 (Network layer).

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 18: ✔ Correct

At which of the following OSI layers does a router operate?

- Layer 1
- Layer 2
- ➡ Layer 3
- Layer 4

EXPLANATION

A router operates at Layer 3, or the Network layer.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 19: ✔ Correct

You are the administrator of your company's network. You want to prevent unauthorized access to your intranet from the internet. Which of the following should you implement?

- Proxy server
- ICS
- Packet Internet Groper
- ➡ Firewall

EXPLANATION

A firewall allows you to filter unwanted traffic from the internet to your network. Packet internet groper is better known by its acronym, PING, a TCP/IP command. A proxy server caches web pages. ICS allows you to connect a small network to the internet through a single connection.

REFERENCES

LabSim for Network Pro, Section 3.3.

Question 20: ✔ Correct

Which device sends signals from a computer onto a network?

- Cable
- Router
- Gateway
- ➡ Transceiver

EXPLANATION

A transceiver (short for transmitter/receiver) sends signals to and receives signals from the network. It translates the parallel data stream of the computer to the serial data stream of the network and vice versa. Most transceivers are now built into network interface cards (NICs).

REFERENCES

LabSim for Network Pro, Section 3.1.

Question 21: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 22: ✔ Correct

Which of the following is the best device to deploy if you want to protect your private network from a public untrusted network?

- Hub
- Gateway
- ➡ Firewall
- Router

EXPLANATION

A firewall is the best device to deploy if you want to protect your private network from a public untrusted network. Firewalls are used to control traffic entering and leaving your trusted network environment. Firewalls can manage traffic based on source or destination IP address, port number, service protocol, application or service type, user account, and even traffic content.

Routers offer some packet-based access control, but it is not as extensive as what a full-fledged firewall provides. Hubs and gateways are not sufficient for managing the interface between a trusted and an untrusted network.

REFERENCES

LabSim for Network Pro, Section 3.3.

Question 23: ✔ Correct

At which layer of the OSI model do hubs operate?

- Internet
- ➡ Physical
- Layer 3
- Data Link

EXPLANATION

Hubs operate at Layer 1, or the Physical layer of the OSI model.

REFERENCES

LabSim for Network Pro, Section 3.2.

Question 24: ✔ Correct

Which of the following is a good reason to install a firewall?

- To translate between internet IP addresses and the IP addresses on your private network.
- ➡ To prevent hackers from accessing your network.
- To limit public access to just your DHCP server instead of your whole network.
- To cache web pages.

EXPLANATION

Firewalls prevent unauthorized users from accessing private networks connected to the internet. You should never allow public access to your DHCP server. A proxy server caches web pages. A NAT router or ICS translates web addresses to private IP addresses. A proxy server and NAT software might be implemented on the same device as your firewall, but they are different concepts.

REFERENCES

LabSim for Network Pro, Section 3.3.

Question 25: ✔ Correct

You are the network administrator for a small organization. Recently, you contracted with an ISP to connect your organization's network to the internet to provide users with internet access. Since doing so, it has come to your attention that an intruder has invaded your network from the internet on three separate occasions.

What type of network hardware should you implement to prevent this from happening again?

- CSU/DSU
- Hub
- ➡ Firewall
- Proxy server
- Router
- Switch

EXPLANATION

The role of a firewall is to provide a barrier between an organization's network and a public network, such as the internet. Its job is to prevent unauthorized access into the organization's private network. To do this, the firewall examines incoming packets and determines whether they should be allowed to enter based on a set of rules defined by the network administrator.

REFERENCES

LabSim for Network Pro, Section 3.3.

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